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Exercise Sheet 10

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Total points: 10

Please upload your solutions to WueCampus as a scanned document (image format or pdf), a typesetted PDF document, and/or as a `jupyter` notebook.

1. Information Theory and Data Compression

(a) Demonstrate the additivity of the information measure by showing that the information gained from observing the collective outcome of N independent events, each with probabilities p_i for $i = 1$ to N, is equal to the sum of the information gained from observing each event individually and independently, regardless of the order in which they are observed. 3P
(Hint: The information gain measure assigns $log_2(p)$ bits to an event with a probability of p.)

Solution: The information measure assigns $log_2(p)$ bits to the observation of an event whose probability is p. The joint probability of a combination of N independent events whose probabilities are $p_1 \dots p_N$ is $\prod_{i=1}^{N} p_i$. Thus the information content of such a combination is:

$$log_2(\prod_{i=1}^{N} p_i) = log_2(p_1) + log_2(p_2) + \dots + log_2(p_N)$$

which is the sum of the information content of all of the separate events.

2. Graphs and Description Length Minimization

(a) Consider a graph $G = (V, E)$ with the following adjacency matrix: 2P

$$\mathbb{A} = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 2 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Assuming that the row and column numbers correspond to nodes (enumerated from 1 to 5), calculate the probability that a random walker starting in node 1 will traverse the following sequence of nodes: $(1, 2, 3, 2, 4, 1, 5, 1, 2)$.
(Hint: Condiser $p(1, 2, 3, 2, 4, 1, 5, 1, 2|1)$.)

Solution:

$$\begin{aligned} p(1,2,3,2,4,1,5,1,2|1) &= \\ = p(2|1)p(3|2)p(2|3)p(4|2)p(1|4)p(5|1)p(1|5)p(2|1) &= \\ = \frac{1}{3}\frac{1}{4}\frac{1}{1}\frac{1}{2}\frac{1}{3}\frac{1}{3}\frac{1}{1}\frac{1}{3} &= \\ &= \frac{1}{648} \end{aligned}$$

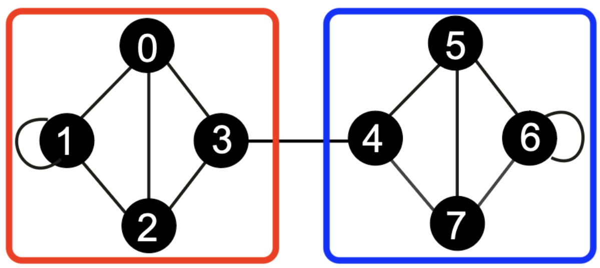
(b) Show that for an undirected network, the stationary visitation probability of a node v converges to $\pi_v = \frac{d(v)}{2|E|}$. 2P

Solution: We can apply the Markov chain convergence theorem. By the Markov Chain Convergence Theorem we now that a stationary distribution exists, meaning that the sequence of states given by $\pi^{i+1} = \pi^i \cdot T$ converges.

$$\pi_v = \sum_{u \in N(v)} \pi_u \cdot T_{vu} = \sum_{u \in N(v)} \frac{d(u)}{2|E|} \cdot \frac{1}{d(u)} = \frac{d(v)}{2|E|}$$

3. **MapEquation**

(a) Consider the following undirected network $G = (V, E)$ and a mapping $C_2 : V \to \{1, 2\}$ of nodes into two communities (red and blue). Use the MapEquation to calculate the minimal code length $L(C_2)$. 3P
(Hint: In this case, assume that a self-loop only contributes 1 to the degree of the node)



Solution: The map equation is given as

$$L(C_k) = qH(Q) + \sum_{i=1}^{k} p_i H(P_i)$$

where q is the per step probability that a random walker switches between communities, Q are the visitation probabilities, p_i is the probability that a random walker visits a node in community i and P_i are the visitation probabilities of nodes in community i.
In our case:

- $q = \frac{2}{24}$ (Random walker switches community if it traverses the edge $(3, 4)$ in one of the two directions. There are 24 possible ways to traverse an edge.)
- $H(Q) = -2 \cdot \frac{1}{2} \log_2 \frac{1}{2} = 1$ since we have the same visitation probability for both groups.
- $p_i = 12/24$ (12 possible ways to traverse an edge and land in community i)
- $H(P_i) = -4 \cdot \frac{1}{4} \log_2 \frac{1}{4} = 2$ (Since all nodes have the same degree and therefore the same visitation probability)

Thus,

$$L(C_2) = \frac{2}{24} \cdot 1 + \frac{12}{24} \cdot 2 + \frac{12}{24} \cdot 2 = \frac{26}{12} \approx 2.083 \text{ bits}$$